

End-to-End Sales Forecasting & Demand Intelligence

Executive Business Report | Prepared for Head of Supply Chain & CFO

1. Executive Summary

We built an intelligent sales forecasting system using 4 years (2015-2018) of Superstore daily transaction data covering multiple product categories and US regions. The system predicts future product demand for the next 3 months, detects unusual sales spikes or drops, segments products by demand behavior, and delivers everything through an interactive dashboard. Our best model (XGBoost) achieves a MAPE of 13.58% after correcting for data leakage, meaning forecasted sales are within 13.58% of actual values on average. We identified 3 distinct product demand clusters with tailored stocking strategies, and detected anomalous weeks driven by seasonal events and promotions.

2. Key Findings from EDA & Forecasting

- **Revenue by Category:** Technology leads with \$827K in total revenue, followed by Furniture (\$729K) and Office Supplies (\$705K). Technology products have the highest average order value.
- **Regional Performance:** West region shows the highest total sales and the most consistent year-over-year sales growth with stable growth rates. East region shows the highest consistency in growth rate trends (YoY growth rate std dev is only 1.8%, compared to Central's 25.3%).
- **Shipping Efficiency:** Average shipping time is ~3.96 days across all regions. East region ships fastest (~3.91 days), followed closely by West (~3.93 days), while Central averages the longest at ~4.07 days.
- **Seasonality:** Strong Q4 spikes (October-December) across all 4 years, driven by holiday demand. January-February consistently show the lowest sales. The seasonal decomposition confirms a 12-month cycle.

3. 3-Month Sales Forecast

We built and compared three fundamentally different forecasting models on monthly sales data:

Model	MAE	RMSE	MAPE (%)	Month 1	Month 2	Month 3
SARIMA	\$19,244	\$19,950	20.53%	\$60,332	\$91,458	\$97,168
Prophet	\$20,296	\$22,487	21.89%	\$51,084	\$90,045	\$89,661
XGBoost	\$14,277	\$19,569	13.58%	\$86,466	\$85,287	\$84,192

Recommendation: XGBoost is recommended for production use — it has the lowest MAPE (13.58%) and the lowest MAE (\$14,277), meaning its forecasts are closest to actual sales on average. Forecasted sales for the next 3 months are in the range of \$84K-\$86K per month, showing strong consistency during the peak Q4 sales period.

4. Top 3 Anomalies Detected

Date	Sales	Type	Likely Cause
2015-03-22	\$37,704	Extreme Spike	Possible corporate bulk order or promotional event (Q1 spike)
2018-12-02	\$35,999	Holiday Spike	Festive season / holiday sales rush (early December peak)
2015-02-22	\$225	Deep Slump	Post-holiday demand contraction (lowest recorded weekly sales)

5. Product Demand Segmentation & Stocking Strategy

A KMeans clustering model categorized 17 sub-categories into 3 distinct business segments:

1. High Volume, Stable Demand (Phones, Chairs, Storage, Binders, Accessories, Tables, Machines)

Strategy: Maintain high safety stock (~14 days), establish automated reorder points to prevent stockouts.

2. Growing Demand (Copiers)

Strategy: Gradually build up inventory, negotiate volume discounts with suppliers to improve margins.

3. Low Volume, Volatile Demand (Art, Fasteners, Paper, Labels, Envelopes, Bookcases, Furnishings, Supplies)

Strategy: Implement Just-In-Time (JIT) replenishment to minimize holding costs, and launch bundle promos to smooth out sales spikes.

6. Actionable Business Recommendations

1. Q4 Supply Chain Preparedness: Because sales surge by over 200% in Q4 (Oct-Dec) compared to Q1, supply chains must preemptively increase stocking levels by late September. Focus safety stock buffers specifically on the High Volume, Stable Demand clusters (Phones, Chairs) to capture peak holiday revenues without capital lockup.

2. Capital Allocations for High-Growth Sectors: Allocate marketing budgets and procurement pipelines to support the Growing Demand cluster (Copiers), which exhibits an exceptional year-over-year trajectory, while scaling down holding capacity for the Low Volume, Volatile group to optimize warehouse efficiency.